Results outline

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1 Wildchrokie

Table 1: **Table Whatever** This table is just produced with somerandom data and does not mean anything. Just to show you how things work.

Model	α	β	t-value	R^2
1	1.96	0.97	0.03	0.49
2	1.95	0.01	0.04	-0.00
3	1.95	0.01	0.04	-0.00

1.1 Growing season length and conditions

1. Across the three years and all species, the growing season length varied from 90 days to 173 days.
2. The year with longest growing season length on average was 2019 with 141.49 days, and the shortest was 2020 with 135.87 days.
3. The hottest spring period of the three years (MAM) was 2020 with a mean mean temperature of 9.05°C, and the coldest, 2019 with a mean mean temperature of 8.72°C.
4. The hottest summer period of the three years (JJA) was 2020 with a mean temperature of 23.07°C, and the coldest, 2019 with a mean temperature of 22.38°C.
5. The driest spring period of the three years (MAM) was 2020 with a PDSI value of 1.32, and the wettest, 2019 with a PDSI value of 2.74.
6. The driest summer period (PDSI) of the three years (JJA) was 2020 with a PDSI value of -2.35, and the wettest, 2019 with a PDSI value of 4.02.
7. The spring period with the highest precipitation of the three years (MAM) was 2018 with a precipitation of 120.87, and the lowest, 2020 with a precipitation of 91.77.
8. The summer period with the highest precipitation of the three years (JJA) was 2019 with a precipitation of 115.33, and the lowest, 2020 with a precipitation of 66.37.

1.2 Phenology observations

1. Leafout dates across all years and species varied from 112 to 157 and budset date varied from 225 to 297.
2. On average, the earliest leafout date was in 2019 for all species (A. incana: 120.86, B alleghaniensis: 123, B. papyrifera: 123.33 and B. populifolia: 121. The latest leafout date for all species was in 2020 (A. incana: 148.68, B alleghaniensis: 141.32, B. papyrifera: 136.57 and B. populifolia: 133.83). No species was consistently the earliest nor the latest one to leafout.

3. On average, the earliest budset date was in 2019 for A. incana (263.29) and B alleghaniensis (258), but in 2018 for B. papyrifera (266.83) and B. populifolia (266.29). The latest budset date for all species was in 2020 (A. incana: 280.88, B alleghaniensis: 264.05, B. papyrifera: 279.43 and B. populifolia: 282). No species was consistently the earliest nor the latest set bud.

1.3 Ring widths

1. The ring width of our species vary from 0.39mm to 13.55mm.

2. Ring widths for A. incana varied from 0.61 to 7.81, from 0.39 to 6.92 for B. alleghaniensis, from 0.65 to 6.42 for B. papyrifera and from 0.75 to 13.55 for B. populifolia.

1.4 Model results

Our results show similar trends in response to the thermal (GDD) versus the calendar growing season (GSL), with a stronger response to longer GSL compared to increased GDD. Especially for A. incana not shifting its growth in response to GDD but showing a strong response to GSL, as well as the start and end of season.

1. A. incana (β = 0.28, UI_{90}[-0.1, 0.65]), B. alleghaniensis (β = 0.18, UI_{90}[-0.24, 0.59]) and B. papyrifera (β = 0.29, UI_{90}[-0.27, 0.83]) show a weak response, but seem to grow more with more GDD, while B. populifolia (β = 0.08, UI_{90}[-0.36, 0.52]) did not respond to increased GDD (Figure 1).

2. A. incana (β = 0.38, UI_{90}[0.14, 0.62]) is the only species for which its 90% intervals don't overlap with zero suggesting that it grows more with longer growing season (GSL). B. alleghaniensis (β = 0.19, UI_{90}[-0.11, 0.5]) and B. papyrifera (β = 0.33, UI_{90}[-0.11, 0.78]) show a weaker response, but seem to grow more in response to longer seasons, while B. populifolia (β = 0.02, UI_{90}[-0.32, 0.36]) did not respond to the GSL (Figure 1).

3. Ring width decreased with a later start of season (SOS) for A. incana (β = -0.66, UI_{90}[-0.9, -0.43]) and B. populifolia (β = -0.45, UI_{90}[-0.83, -0.05]). Whereas B. alleghaniensis (β = -0.2, UI_{90}[-0.55, 0.14]) and B. papyrifera (β = -0.21, UI_{90}[-0.69, 0.27]) show a weaker response, but seem to grow less with later SOS. (Figure 1).

4. Ring width decreased with a later end of season (EOS) for A. incana (β = -0.27, UI_{90}[-0.52, -0.02]). B. populifolia showed a weaker response but appears to also grow less with a later EOS (β = -0.25, UI_{90}[-0.54, 0.05]). B. alleghaniensis (β = 0.04, UI_{90}[-0.25, 0.33]) and B. papyrifera (β = 0.06, UI_{90}[-0.33, 0.45]) didn't respond to EOS. (Figure 1).

5. For the provenance with the GDD model, the estimates didn't shift much across climate predictors. We detected a weak growth difference between G. College (α_{site} = -0.33, UI_{90}[-1.96, 1.27]), S. Hypolyte (α_{site} = -0.03, UI_{90}[-1.7, 1.6]) or W. Mountains (α_{site} = 0.51, UI_{90}[-1.17, 2.09]). Only HF grew less (α_{site} = = -0.72, UI_{90}[-2.39, 0.92]) (Figure 1)

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Supplemental results

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1. Fitting start of season (leafout) adds 12 rows of data out of the 227 when removing both NA entries for leafout and budset. Fitting end of season (budset) adds 3 rows of data out of the 227 when removing both NA entries for leafout and budset.

2. Fitting with the maximum number of rows of data didn't change the parameter estimates by much for SOS (Figure S1) and EOS (Figure S2).

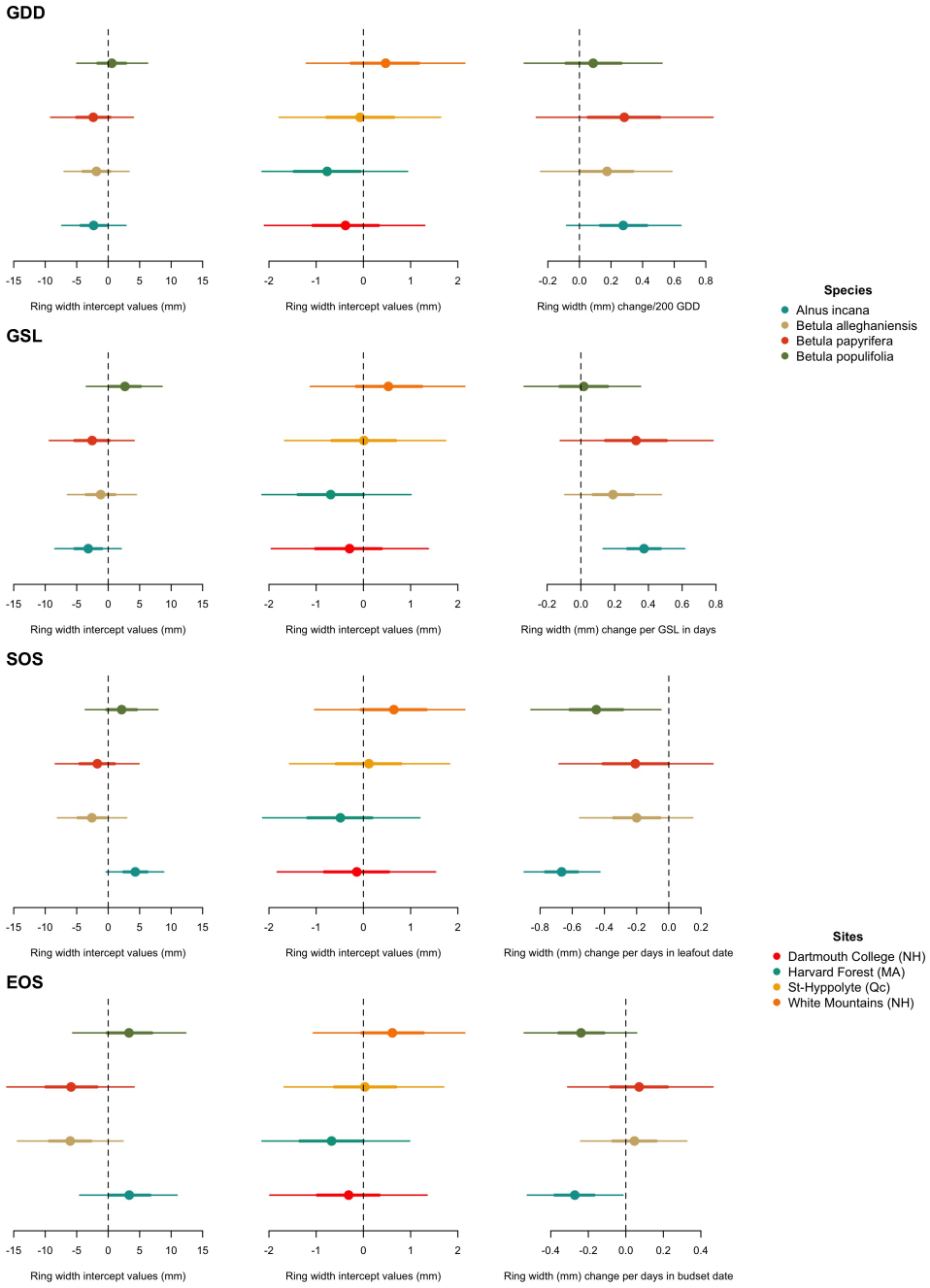

Figure 1: Model parameters across the different predictors

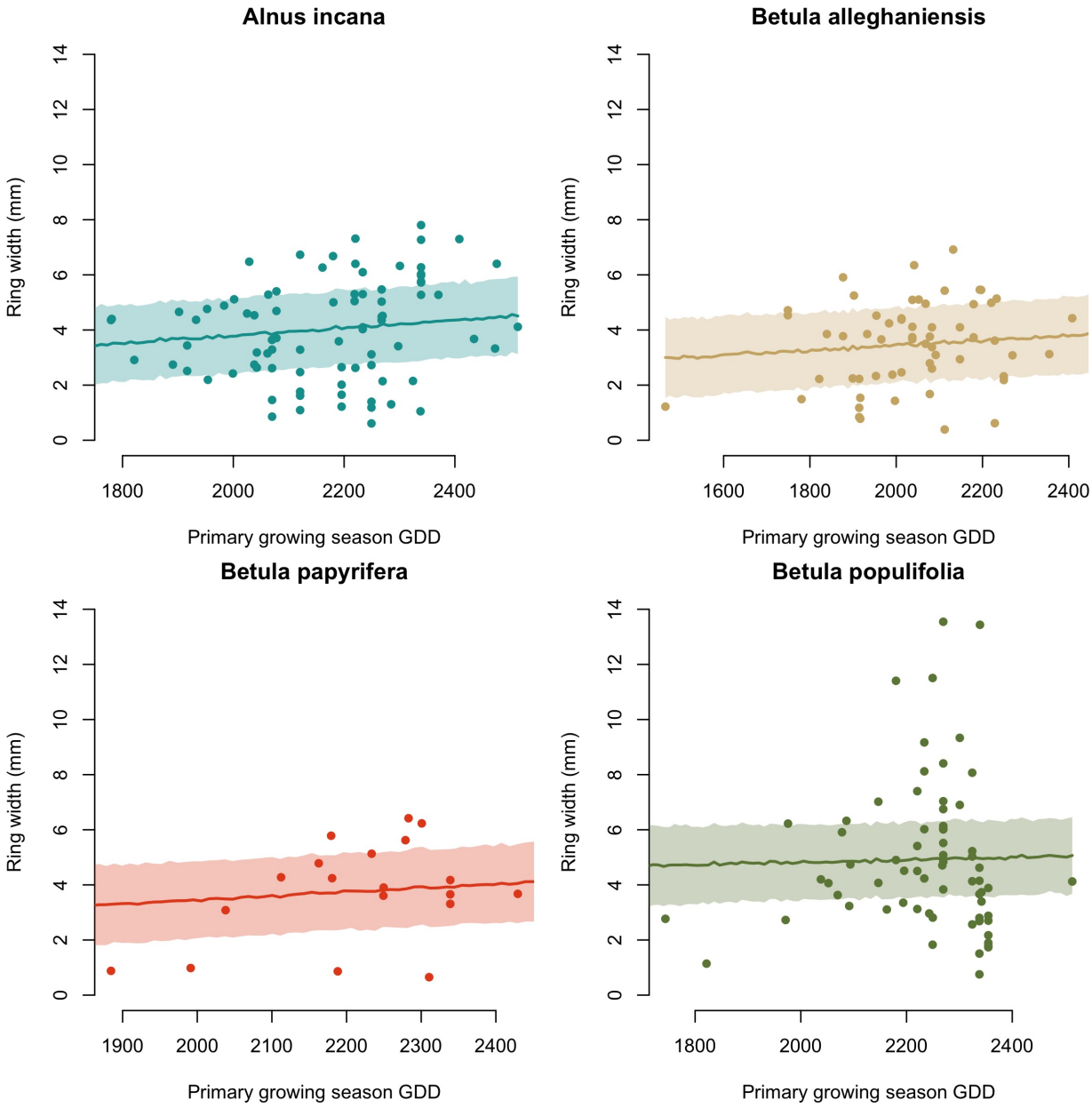

Figure 2: Ring width responses to growing degree days for the primary growing season. The shaded areas represent the 50% confidence intervals. The dots represent the empirical data

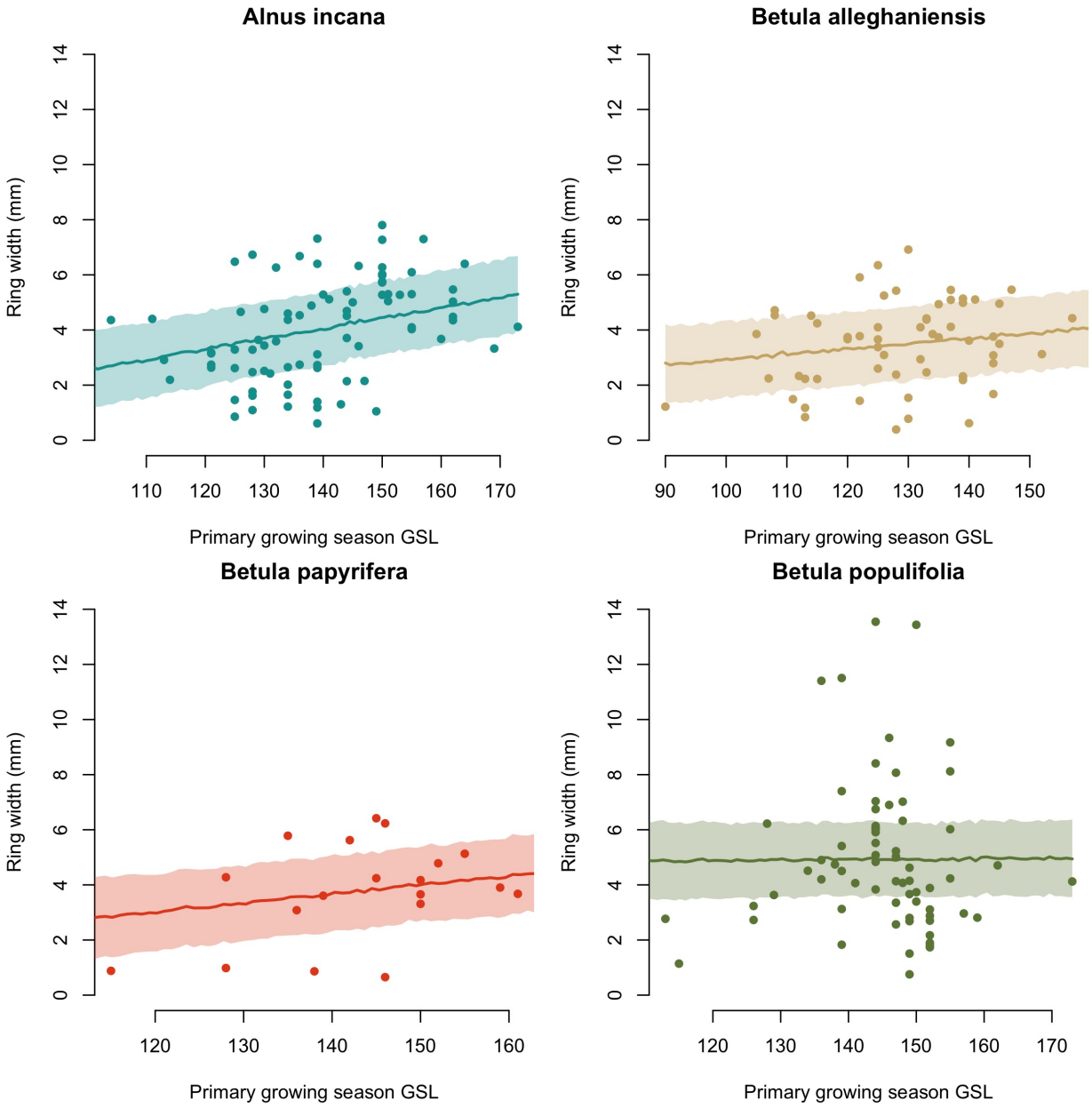

Figure 3: Ring width responses to length of the primary growing season (days). The shaded areas represent the 50% confidence intervals. The dots represent the empirical data

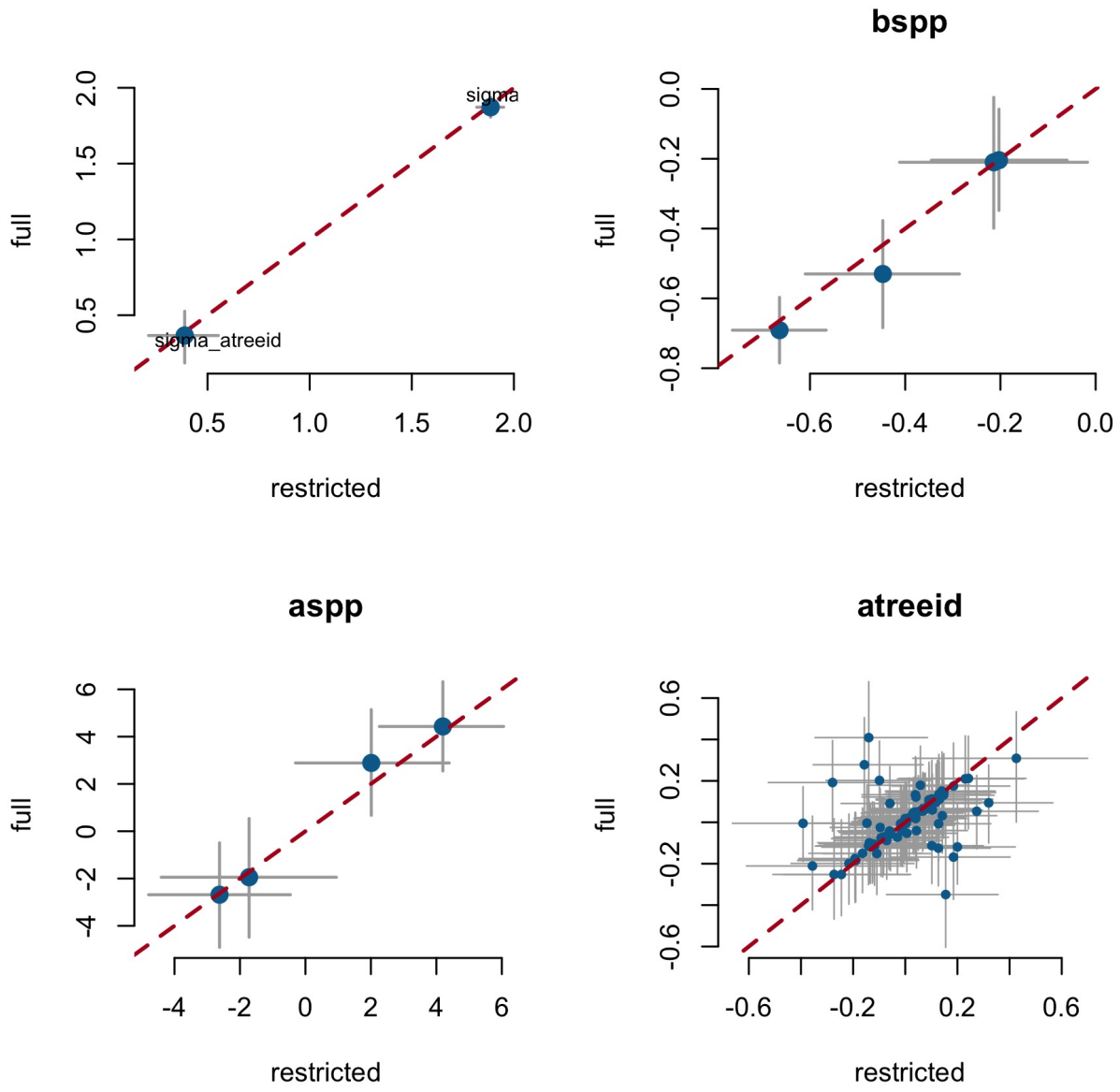

Figure S1: Full vs most restricted rows of data for the model with SOS as the predictor.

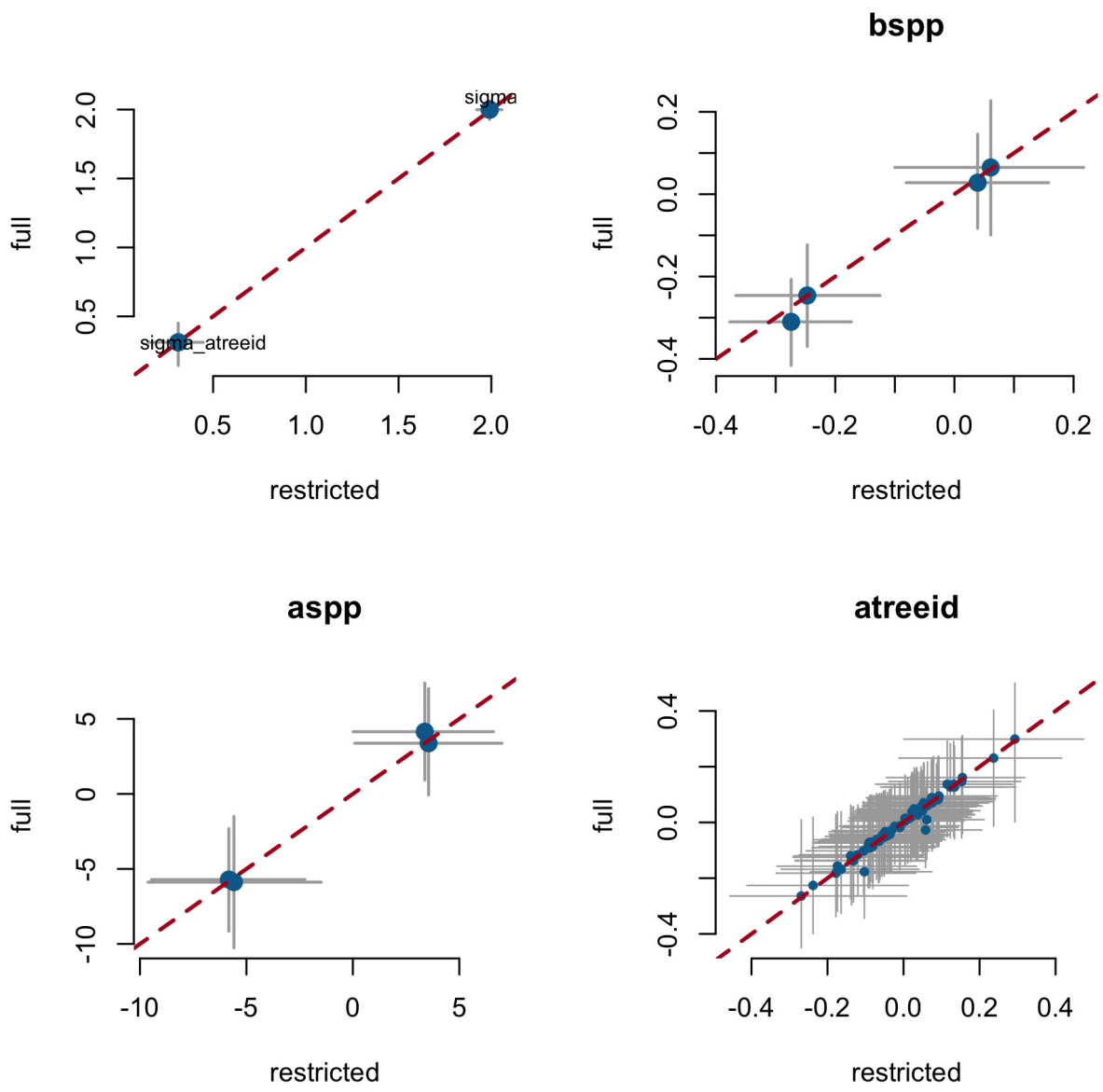

Figure S2: Full vs most restricted rows of data for the model with EOS as the predictor.